

CO 3 : ASSESSMENT TOOL 3 :

CODING CHALLENGE QUESTIONS

- RAGAVI S
192565027

QUESTION 1 –

INPUT :

The screenshot shows an 8085 assembly simulator. On the left, the assembly code is displayed with line numbers 1 through 27. The code includes instructions like MVI, LXI, MOV, INX, CMP, JZ, JNC, DCR, and HLT. On the right, the 'Memory' tab is active, showing a memory dump starting at address 8050. The dump lists addresses from 1F72 to 1F7E with their corresponding data values.

Address (Hex)	Address	Data
1F72	8050	4
1F73	8051	8
1F74	8052	3
1F75	8053	7
1F76	8054	1
1F77	8055	9
1F78	8056	6
1F79	8057	2
1F7A	8058	5
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0

OUTPUT :

This screenshot shows the same 8085 assembly simulator after execution. The assembly code on the left is identical to the input. The 'Memory' tab on the right shows a different memory dump, reflecting the changes made by the program. The data values at addresses 8050 through 8061 are now different. At the bottom, a status message indicates that the program was assembled successfully.

Address (Hex)	Address	Data
1F72	8050	1
1F73	8051	2
1F74	8052	3
1F75	8053	4
1F76	8054	6
1F77	8055	7
1F78	8056	8
1F79	8057	9
1F7A	8058	0
1F7B	8059	0
1F7C	8060	1
1F7D	8061	0

0 Program assembled successfully

QUESTION 2 –

INPUT :

Load me at

```
1 LXI H,8000
2 LXI D,8100
3 MVI C,08
4
5 COPY: MOV A,M
6 STAX D
7 INX H
8 INX D
9 DCR C
10 JNZ COPY
11
12 LXI H,8000
13 LXI D,8100
14 MVI C,08
15 MVI B,00
16
17 COMPARE: LDAX D
18 CMP M
19 JZ SAME
20 INR B
21
22 SAME: INX H
23 INX D
24 DCR C
25 JNZ COMPARE
26
27 MOV A,B
28 STA 8200
29 HLT
```

Data Stack KeyPad Memory I/O Ports

Start 8000 OK

Address (Hex)	Address	Data
1F40	8000	11
1F41	8001	22
1F42	8002	33
1F43	8003	44
1F44	8004	55
1F45	8005	66
1F46	8006	77
1F47	8007	88
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0

Line No Assembler Message

Load me at

```
1 LXI H,8000
2 LXI D,8100
3 MVI C,08
4
5 COPY: MOV A,M
6 STAX D
7 INX H
8 INX D
9 DCR C
10 JNZ COPY
11
12 LXI H,8000
13 LXI D,8100
14 MVI C,08
15 MVI B,00
16
17 COMPARE: LDAX D
18 CMP M
19 JZ SAME
20 INR B
21
22 SAME: INX H
23 INX D
24 DCR C
25 JNZ COMPARE
26
27 MOV A,B
28 STA 8200
29 HLT
```

Data Stack KeyPad Memory I/O Ports

Start 8100 OK

Address (Hex)	Address	Data
1FA4	8100	0
1FA5	8101	0
1FA6	8102	0
1FA7	8103	0
1FA8	8104	0
1FA9	8105	0
1FAA	8106	0
1FAB	8107	0
1FAC	8108	0
1FAD	8109	0
1FAE	8110	0
1FAF	8111	0
1FER0	8112	0

Line No Assembler Message

OUTPUT :

Load me at

```
1 LXI H,8000
2 LXI D,8100
3 MVI C,08
4
5 COPY: MOV A,M
6 STAX D
7 INX H
8 INX D
9 DCR C
10 JNZ COPY
11
12 LXI H,8000
13 LXI D,8100
14 MVI C,08
15 MVI B,00
16
17 COMPARE: LDAX D
18 CMP M
19 JZ SAME
20 INR B
21
22 SAME: INX H
23 INX D
24 DCR C
25 JNZ COMPARE
26
27 MOV A,B
28 STA 8200
29 HLT
```

Data Stack KeyPad Memory I/O Ports

Start 8100 OK

Address (Hex)	Address	Data
1FA4	8100	11
1FA5	8101	22
1FA6	8102	33
1FA7	8103	44
1FA8	8104	55
1FA9	8105	66
1FAA	8106	77
1FAB	8107	88
1FAC	8108	0
1FAD	8109	0
1FAE	8110	0
1FAF	8111	0

Line No Assembler Message

0 Program assembled successfully

QUESTION 3 –

INPUT :

Load me at

```
1 LDA 8050
2 MOV B,A
3 MVI C,08
4 MVI D,00
5
6 LOOP: MOV A,B
7 ANI 01
8 JZ SKIP
9 INR D
10
11 SKIP: MOV A,B
12 RRC
13 MOV B,A
14 DCR C
15 JNZ LOOP
16
17 MOV A,D
18 ANI 01
19 JZ EVEN
20
21 MVI A,00
22 STA 8052
23 HLT
24
25 EVEN: MVI A,01
26 STA 8052
27 HLT
```

Data Stack KeyPad Memory I/O Ports

Start 8050 OK

Address (Hex)	Address	Data
1F72	8050	24
1F73	8051	0
1F74	8052	0
1F75	8053	0
1F76	8054	0
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0

Line No Assembler Message

OUTPUT :

Load me at

```
1 LDA 8050
2 MOV B,A
3 MVI C,08
4 MVI D,00
5
6 LOOP: MOV A,B
7 ANI 01
8 JZ SKIP
9 INR D
10
11 SKIP: MOV A,B
12 RRC
13 MOV B,A
14 DCR C
15 JNZ LOOP
16
17 MOV A,D
18 ANI 01
19 JZ EVEN
20
21 MVI A,00
22 STA 8052
23 HLT
24
25 EVEN: MVI A,01
26 STA 8052
27 HLT
```

Data Stack KeyPad Memory I/O Ports

Start 8050 OK

Address (Hex)	Address	Data
1F72	8050	24
1F73	8051	0
1F74	8052	1
1F75	8053	0
1F76	8054	0
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0

Line No Assembler Message

0 Program assembled successfully